

Liangliang Zhang

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EDUCATION

Rensselaer Polytechnic Institute (RPI)

Ph.D. in Computer Science, advised by Prof. Yao Ma

Troy, NY

Aug. 2024 – May 2028 (Expected)

Central South University

B.S. in Software Engineering

Changsha, China

Sep. 2020 – May 2024

RESEARCH INTERESTS

Retrieval-Augmented Generation, Agentic Retrieval and Reasoning, Knowledge-Grounded LLMs, and Reliable Evaluation.

RESEARCH EXPERIENCE

IBM Research

Research Intern

Yorktown Heights, NY

May 2026 – Aug. 2026

- Developed AgenticTT, an agentic framework for adaptive retrieval and reasoning over heterogeneous tables and text, enabling LLM agents to dynamically search, navigate, operate on evidence, and answer.
- Designed a typed evidence interface and reliability controls for grounded multi-step execution across tables, rows, passages, and fused table-text evidence.
- Achieved 75.8 EM on OTT-QA, 68.2 EM on LakeQA-mini, and 70.6 EM on HybridQA, including gains of +16.8 EM and +43.8 EM over matched baselines on OTT-QA and LakeQA-mini.
- Resulted in “Adaptive Agentic Retrieval and Reasoning for Table-Text QA,” under review at AAAI 2027.

IBM Research

Research Intern

Yorktown Heights, NY

May 2025 – Aug. 2025

- Built a benchmark with 13K+ data product requests for heterogeneous table-text discovery across Wikipedia and financial domains, extending factoid retrieval to multi-source evidence discovery.
- Evaluated BM25, dense, hybrid, and reranking retrieval and identified candidate generation and multi-source evidence coverage as key bottlenecks in heterogeneous retrieval.
- Resulted in “From Factoid Questions to Data Product Requests: Benchmarking Data Product Discovery over Tables and Text” (arXiv:2510.21737).

DAMI Lab, Rensselaer Polytechnic Institute

Graduate Research Assistant, advised by Prof. Yao Ma

Troy, NY

Aug. 2024 – Present

- Served as Teaching Assistant for **CSCI 4150: Introduction to Artificial Intelligence** in Spring 2026, leading laboratory sessions and office hours and evaluating assignments and examinations.
- Served as Teaching Assistant for **CSCI 1100: Computer Science** in Fall 2024, leading laboratory sessions and office hours and evaluating assignments and examinations.

Institute of Automation, Chinese Academy of Sciences

Research Intern

Beijing, China

May. 2023 – Sep. 2023

- Implemented and evaluated adversarial robustness methods for deep neural networks, including **CW**, **AutoAttack**, and **CopyCAT**, using PyTorch, Linux, and Docker.

CONFERENCE AND JOURNAL PUBLICATIONS

- [1] **Zhang, Liangliang**, Z. Jiang, H. Chi, H. Chen, M. Elkoumy, F. Wang, Q. Wu, Z. Zhou, S. Pan, S. Wang, et al. “Diagnosing and Addressing Pitfalls in KG-RAG Datasets: Toward More Reliable Benchmarking”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2025.
- [2] **Zhang, Liangliang**, H. Bao, and Y. Ma. “Extending Graph Condensation to Multi-Label Datasets: A Benchmark Study”. In: *Transactions on Machine Learning Research (TMLR)* (2025).
- [3] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and J. Bu. “Learning to Reduce the Scale of Large Graphs: A Comprehensive Survey”. In: *ACM Transactions on Knowledge Discovery from Data* 19.5 (2025), pp. 1–25.
- [4] Zhao, Tianyi*, **Zhang, Liangliang***, Y. Ma, and L. Cheng. “A Survey on Safe Multi-Modal Learning System”. In: *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. *Equal contribution. 2024.
- [5] **Zhang, Liangliang**, M. Chen, X. Hu, and L. Deng. “Graph Convolutional Network and Contrastive Learning Small Nucleolar RNA (snoRNA) Disease Associations (GCLSDA): Predicting snoRNA–Disease Associations via Graph Convolutional Network and Contrastive Learning”. In: *International Journal of Molecular Sciences* 24.19 (2023), p. 14429.
- [6] D. Liu, X. Li, **Zhang, Liangliang**, X. Hu, J. Zhang, Z. Liu, and L. Deng. “HGNNLDA: Predicting lncRNA-Drug Sensitivity Associations via a Dual Channel Hypergraph Neural Network”. In: *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 20.6 (2023), pp. 3547–3555.

PREPRINTS

- [1] **Zhang, Liangliang**, N. Mihindukulasooriya, N. S. D’Souza, S. Shirai, S. Dash, Y. Ma, and H. Samulowitz. “From Factoid Questions to Data Product Requests: Benchmarking Data Product Discovery over Tables and Text”. In: *arXiv preprint arXiv:2510.21737* (2025).
- [2] H. Xu, **Zhang, Liangliang**, Y. Ma, S. Zhou, Z. Zheng, and B. Jiajun. “A Survey on Graph Condensation”. In: *arXiv preprint arXiv:2402.02000* (2024).

ADDITIONAL TECHNICAL EXPERIENCE

Hunan Wuche Education Technology Co., Ltd.

China

Software Project Intern

Aug. 2022 – Sep. 2022

- Developed an online tank battle game using **JavaFX** and **FXGL**, integrating user interfaces, application logic, database components, and email functionality.

China Hi-Tech Group

China

Software Project Intern

Jun. 2022 – Jul. 2022

- Developed a full-stack shopping mall application using **Vue** and **Spring Boot**, covering front-end interfaces and supporting back-end services.

Guangdong QST Green Soft Training Education Technology Co., Ltd.

China

Software Project Intern

Jun. 2021 – Jul. 2021

- Developed an online Gomoku game in **C++**, implementing user authentication, multiple game modes, and core gameplay logic.

SELECTED HONORS & AWARDS

- **Best Doctoral Consortium Presentation Award**, ACM WSDM 2026 Doctoral Consortium.
- **Academic Year Scholarship**, Central South University — awarded four times.

TECHNICAL SKILLS

Programming: Python, C++, SQL, Java **ML/LLMs:** PyTorch, PyTorch Geometric, Hugging Face Transformers, vLLM

Retrieval / Data Systems: FAISS, OpenSearch, Milvus, BM25, Dense Retrieval, Reranking

Tools: Linux, Docker, Git